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Education

San Jose State University

Master of Science in Software Engineering

Jan. 2025 - Dec. 2026

San Jose, CA

SRM Institute of Science and Technology

Bachelor of Technology in Computer Science and Engineering (Specialization in Cloud Computing)

Aug. 2020 - May 2024

Chennai, India

Experience

Microsoft

May 2026 - Aug. 2026

Software Engineering Intern, Security Platform & AI (SP&AI)

Redmond, WA

- Productionized **FluxBridge**, a multi-tenant security data-lake ingestion pipeline on **AKS**, enabling **Vector** workers to process protobuf-encoded security telemetry into **Parquet** for SentinelDataLake/ADLS across **50–80** input streams.
- Built an end-to-end **OpenTelemetry** telemetry plane for Vector workers, reshaping **internal_metrics** and **internal_logs** into **OTLP** with VRL and tagging every signal with **worker.id**, **proto.schema**, and service metadata for per-stream debugging.
- Deployed an in-cluster **OpenTelemetry Collector** gateway on AKS, forwarding OTLP telemetry to **App Insights**, **Log Analytics**, and Azure **Grafana**; validated **98K+ AppMetrics rows/hour** and delivered **8** operational dashboard panels.
- Hardened the AKS deployment for compliance-constrained environments by packaging the **C# IngestionWorker**, **Rust**-patched **Vector** binary, and pipeline configs into an **ACR**-hosted image aligned with per-stream autoscaling.
- Productionized a custom **Go** metrics-action component that consumes Vector runtime metrics from **Prometheus**, enabling configurable remediation actions based on throughput, buffer depth, drain state, and error-growth thresholds.
- Engineered a **C#** drain-detection mechanism that scrapes Vector's **Prometheus /metrics** endpoint, correlating sent events, buffered events, and error growth to detect settled sinks and eliminate indefinite worker hangs caused by internal metrics source.

AWS Cloud Virtual Internship - AWS Academy

May 2023 - July 2023

Cloud Intern

Remote

- Architected a secure **2-tier AWS deployment** with **EC2**, **S3**, **VPC**, and **Route 53**, implementing subnet isolation, route-table segmentation, DNS routing, and security-group controls to separate public entry points from private application resources.
- Improved infrastructure security, performance, and cost posture by hardening **IAM** access, validating network rules, load-testing EC2 configurations, and applying utilization-based right-sizing, reducing over-permissioned access by ~40%, response time by ~20%, and projected cost by ~15%.

Projects and Licenses

cudfgrep: GPU-Accelerated Grep for NVIDIA RAPIDS cuDF | *Python, cuDF, CUDA, Pytest, CLI*

July 2026

- Authored an open-source **NVIDIA RAPIDS cuDF** feature implementing **cudfgrep**, a Unix grep-style CLI that uses cuDF's GPU string engine for massively parallel regex search over text files.
- Implemented grep-compatible functionality including case-insensitive search, line numbers, counts, inverted matches, only-matching output, word/line matching, multi-pattern search, stdin, multi-file prefixes, benchmark mode, and correct grep exit codes.
- Benchmarked on an **NVIDIA T4 GPU**, reaching ~**20 GB/s** GPU-resident regex scan throughput and ~**8–9x** faster scan performance than ripgrep, with correctness verified against CPU grep and ~**30** pytest tests.

Production-Ready Cloud Infrastructure Deployment | *AWS, Terraform, Docker, CloudWatch*

March 2026

- Containerized and deployed backend APIs on AWS using Docker and Terraform, implementing CI/CD pipelines.
- Designed secure VPC architecture with IAM roles, security groups, and load balancing for controlled public access.
- Reduced deployment time by ~75% through reusable Terraform modules and automated provisioning workflows.
- Enabled CloudWatch monitoring and centralized logging, improving fault detection time by ~30% during load testing.

Efficient Task Offloading in Mobile Edge Computing | *Team Lead - AWS, iFogSim, Python*

April 2024

- Designed and implemented a mobile edge computing simulation framework using iFogSim to evaluate task offloading between edge devices and cloud nodes.
- Evaluated latency and energy trade-offs in offloading strategies using real-world cloud scenarios.
- Developed an interactive UI and backend database for task scheduling and monitoring.
- Improved resource allocation efficiency by 25% through adaptive offloading mechanisms.

AWS Academy Cloud Practitioner License (*View Credential*)

August 2023

AWS Academy Cloud Architecting License (*View Credential*)

June 2023

Technical Skills

Languages: Python, Java, C++, C#, Go, Rust, JavaScript, TypeScript, SQL, HTML/CSS, Bash

Cloud & Infrastructure: Azure, AKS, Kubernetes, AWS, EC2, S3, IAM, VPC, Route 53, Azure Monitor, ACR, Terraform, Docker, Jenkins, GitHub Actions, CI/CD

Backend & Observability: OpenTelemetry, OTLP, Vector, Prometheus, Grafana, App Insights, Log Analytics, KQL, REST APIs, Linux/Unix, Git, GitHub, Spring Boot, Flask, Node.js, Express.js

Data & Systems: NVIDIA cuDF, CUDA, Protobuf, Parquet, MySQL, PostgreSQL, MongoDB, Redis, Kafka, pytest, xUnit, Data Structures and Algorithms

Concepts: Distributed Systems, Object-Oriented Design, Cloud Architecture, Microservices, API Development, Performance Benchmarking, Secure Access Control, Agile Development